

Lesson 9: Sensors and circuit choices

Electric Circuits · Year 10 · 50 minutes · AQA 6.2.1.4, 6.2.2

Why this lesson is here

An automatic night light and a temperature alarm use resistors whose resistance responds to the surroundings.

Learning objective

Predict how light and temperature change current in simple sensor circuits.

Core teaching point

NTC thermistor: temperature up, resistance down. LDR: light intensity up, resistance down. At fixed supply p.d., lower total resistance means higher current.

Equipment and preparation

LDR, thermistor and multimeter for demo; graph cards; optional virtual circuit for series response.

Lesson sequence

Time	Phase	Teacher move and evidence
0-5	Retrieval	Recall lamp/resistor/diode characteristics.
5-17	Model sensors	Teach NTC and LDR trends, then circuit consequence.
17-31	Interpret	Groups match contexts, graphs and components.
31-43	Calculate	Tasks 2-4 with fixed supply p.d.
43-50	Check	Four-mark LDR question.

Workbook task guidance

Task 1: Sketch R against temperature for an NTC thermistor and R against light intensity for an LDR.

Expected: Both resistance curves fall as x variable rises; label axes and units/context.

Task 2: Explain the current change in a series circuit containing a thermistor when it gets warmer.

Expected: Warmer NTC thermistor has lower R; total R and series current change accordingly at fixed V.

Task 3: Calculate I for a 12 V sensor circuit with total R = 300 ohm, then 600 ohm.

Expected: 0.040 A then 0.020 A.

Task 4: Choose thermistor or LDR for a fire warning and a streetlamp; justify each choice.

Expected: Thermistor for temperature/fire warning; LDR for light/streetlamp.

Worked example

An LDR circuit has 6.0 V and total R = 600 ohm in bright light: $I = 6.0/600 = 0.010$ A. In dark R = 1200 ohm: $I = 0.0050$ A.

AQA-style checkpoint

[4 marks] A 6.0 V supply is connected to an LDR and a fixed resistor in series. In dim light the total resistance is 1500 ohm. Calculate current. Explain what happens to current when the light becomes brighter.

Mark guidance: $I = V/R$ (1) = $6.0/1500 = 0.0040$ A (1). LDR resistance decreases (1), so total R decreases and current increases (1).

Anticipated misconception

A thermistor's effect here is the NTC type: warmer means less resistance.

Assessment and next step

Collect one sentence of reasoning and one complete calculation or graph label. Start the next lesson by reteaching the most common error; students who are secure should tackle a changed numerical context.